amazon rme

AMAZON BASE BUILDING MAINTENANCE STANDARD

GLOBAL CENTRAL RME STANDARDS & COMPLIANCE

Task Title	HVAC SYSTEMS, AIR HANDLING UNITS - 9
APM Object	HVAC.AHU
APM EC	AMZ.HVAC.AHU.GRP
BBM Handbook Title	Air Handling Units (AHUs)
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure These tasks are based on industry guidance, SME knowledge and HIE learning. Please ensure vendor specific requirements are followed. To suggest improvements please raise a SIM SIM - Link to SIM template

Last Update: (Version)	Aug-23	**Geo Applied:**	Intl APM Instance	**F-Gas**	Yes	**Activies may be covered by No-Go restrictions:**	YES	**Special Certificaion Required:**	Hygiene certification required in some countries (e.g. VDI 6022)	**Delivered By:**	Carried out by trained on site technician certified in the electrical field	**OCB:**	BBM
Evidence for CT:	TBC	**Pricing template Position:**	9	**Related CT Areas:**	CT.AIR.INDOOR			**Assets included in this PM:**	AIR HANDLER UNIT	**Schedule With:**	HVAC.HMD HVAC.BLR (if installed)		

Description of System / Technology (According to BBM Handbook)

An air handler, or air handling unit (AHU), is a device used to regulate and circulate air that is a typical of HVAC system types seen across Amazon FC's. An AHU typically contains the fans, filter chambers and dampers. Heating elements can be integrated (I.e. gas burners) or separate via a DX heating/cooling unit for example. AHU's can also provide free cooling.

Additional Information

All sites with the above listed Base Building Asset should conduct routine maintenance and inspections according to this standard PM package. The Amazon maintenance standard supports compliance, reduces safety risks,improves asset reliability & efficiency.
Some maintenances captures several assets within one PM package -> Ensure that corrective/reactive maintenance for assets where proactive maintenance is contained under another asset is created against the original asset.
Note: the yearly PPM should be planned before the summer season starts.

Task Index	Component	Tasks	Tasks Nested for frequencies > weekly									
			Daily	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	36 Monthly	48 Monthly	60 Monthly
1	General	Is the technical documentation & blueprints of installed system available & complete						PM		PM		
2	AHU outside check	Is the unit freely accessible & maintainable			SC	SC	SC	PM		PM		
3	AHU outside check	Is the unit in operation & no alarms indicated - check unit control cabinet			SC	SC	SC	PM		PM		
4	AHU outside check	Check unit air intakes & outlets for corrosion & damage, contamination & condensation			SC	SC	SC	PM		PM		
5	AHU outside check	Is there any unusual noise or vibration that could be caused by a faulty unit component			SC	SC	SC	PM		PM		
6	AHU outside check	Are there any visual damage, missing- or loose components at the unit			SC	SC	SC	PM		PM		
7	AHU outside check	Is water dropping out of the unit			SC	SC	SC	PM		PM		
8	AHU outside check	Check of the drain & drain trap (e.g. ball siphon) connected to the unit, chek for correct function & damage						PM		PM		
9	AHU filter	Check filters for unacceptable contamination, damage, leaks (can the air bypass the filter medium?)			SC	SC	SC	PM		PM		
10	AHU filter	Measure differential pressure of all filter stages by using adequate tools			SC	SC	SC	PM		PM		
11	AHU filter	List the differential pressures in the notes			SC	SC	SC	PM		PM		
12	AHU filter	Schedule filter change when 80% of the capacity is reached			SC	SC	SC	PM		PM		
13	AHU filter	(An increased filter change is ok due to environmental impact)						PM		PM		
14	AHU filter	In the notes, please enter the number of filter changes per filter stage within the year						PM		PM		
15	AHU Fan	Change belts and check pulleys at belt driven fans						PM		PM		
16	AHU Hygiene & damages	Check droplet eliminator and tray for contamination e.g. mold, corrosion & damage				SC	SC	PM		PM		
17	AHU Hygiene & damages	Check cooling coil & condensate tray for contamination, corrosion & damage				SC	SC	PM		PM		
18	AHU Hygiene & damages	Clean unit air intakes and outlets						PM		PM		
19	AHU Hygiene & damages	Clean droplet eliminator & tray and check for contamination e.g. mold, corrosion & damage						PM		PM		
20	AHU Hygiene & damages	Clean cooling- & heating coil, as well as condensate tray using professional cleaning materials						PM		PM		
21	AHU Hygiene & damages	Check for contamination, corrosion & damage						PM		PM		
22	AHU Hygiene & damages	Clean fans & impeller and check for contamination, corrosion & damage						PM		PM		
23	AHU Hygiene & damages	Clean Chamber surfaces inside the unit and check for contamination, corrosion & damage						PM		PM		
24	AHU Hygiene & damages	Visual check of heat recovery for contamination, corrosion & damage					SC	PM		PM		
25	AHU Hygiene & damages	Visual check of heating coil for contamination, corrosion & damage					SC	PM		PM		
26	AHU Hygiene & damages	Visual check of silencers for contamination, corrosion & damage						PM		PM		
27	AHU Heat exchange	Functional check of frost protection thermostat						PM		PM		
28	AHU Heat exchange	Check the electrical heater for contamination, corrosion & damage						PM		PM		
29	AHU Heat exchange	Check the electrical heater according to local electrical regulations						PM		PM		
30	AHU Heat exchange	Check directly heated heat exchanger for tightness						PM		PM		
31	AHU Heat exchange	Vacuum all remaining heat exchanger carefully						PM		PM		
32	AHU Heat exchange	Check of reheater & cooler installed in the duct system for contamination, corrosion & damage						PM		PM		
33	AHU Heat exchange	Change filter of reheater & cooler installed in the duct system						PM		PM		
34	AHU Heat exchange	Measure differential pressure of all heat exchanger, compare with design and schedule professional cleaning if differential pressure is 30% higher than designed -> use professional tool								PM		
35	AHU Heat Recoery wheel	Check the heat recovery wheel for correct functionality & for damage						PM		PM		

36	AHU duct system	Check air outlets & intakes of the duct system for debris, contamination, corrosion & damage						PM		PM			
37	AHU duct system	Choose 5 locations and add the locations into the notes Check inner duct system checks for contamination, corrosion, condensation & dirt						PM		PM			
38	AHU duct system	Check Textile air hose for damage, contamination & dirt						PM		PM			
39	AHU duct system	Schedule exchange or washing of the textile air hose if necessary						PM		PM			
40	AHU duct system	Underground components - check for contamination, corrosion & dirt deposit inside of the duct						PM		PM			
41	Control cabinet	Visual check of the control cabinets inside & outside for damage, debris & missing components						PM		PM			
42	Control cabinet	Check circuit diagrams for completion						PM		PM			
43	Control cabinet	If anything is missing or unclear in the circuit diagrams create a WO						PM		PM			
44	Control cabinet	Are all cables properly routed						PM		PM			
45	Control cabinet	Are all cables, terminal blocks & devices properly labeled or marked						PM		PM			
46	Control cabinet	Test all safety relevant switches						PM		PM			
47	Control cabinet	Were any obvious changes made (e.g. rewiring, hanging cables)						PM		PM			
48	Control cabinet	Visual check for the protection against direct contact with live part						PM		PM			
49	Control cabinet	Tight cable connections in the control cabinets & main supply connection boxes						PM		PM			
50	Control cabinet	Are all metal parts connected to the protection earth (ground)						PM		PM			
51	Control cabinet	Replace the air filters of the cabinet & check operation of cooling fan						PM		PM			
52	Measurements	Measure the power consumption of the Fans in full speed with open dampers						PM		PM			
53	Measurements	Add the values into the notes indicating the fans & compare to design specifications of the fan						PM		PM			
54	Measurements	Measure the main discharge & return air volume flow - add airflow and position into the notes						PM		PM			
55	Measurements	Compare measured airflow with design specification						PM		PM			
56	Measurements	Measure the main discharge & return air temperature						PM		PM			
57	Measurements	Compare measured temperature with value in the BMS, add temperature and create a FWO under BMS in case of deviation						PM		PM			
58	Final Check	Check the O&M manual and carry out additional required tasks						PM		PM			

Change Log

July 2023 - First issue in new format. Simplification of task scope, alignment with CT.
June 2024 - Moved "Humidifier tasks" to HVAC.HMD PM.